



Incident report analysis

Summary	The organization experienced a Denial of Service (DoS) attack that disrupted normal business operations for two hours by flooding the internal network with ICMP ping requests. The root cause was an unconfigured firewall that allowed malicious traffic to enter and overwhelm network services. The incident was mitigated by blocking incoming ICMP packets, taking non-critical services offline, and restoring core network functions.
Identify	Attack Type: Denial of Service (DoS) attack via ICMP flood (ping flood). Vulnerability: An unconfigured firewall that permitted unrestricted inbound ICMP traffic. Affected Assets & Systems: Internal network infrastructure, network services, and internal user devices attempting to access network resources. Impact: Interruption of normal network traffic and unavailability of internal network resources for two hours.
Protect	To prevent similar incidents, the following controls and configuration updates must be implemented: • Firewall Rules: Configure rate-limiting rules on the firewall to restrict the volume of incoming ICMP requests. • Spoofing Prevention: Implement source IP address verification at the perimeter firewall to filter out spoofed IP addresses attempting to bypass filters. • Firewall Hardening Policies: Establish regular firewall policy audits and baseline configurations to ensure no port or protocol is left unconfigured. • Access & Privilege Controls: Restrict firewall administrative rights strictly to authorized network engineers using Role-Based Access Control (RBAC).
Detect	To ensure rapid identification of potential threats or unusual activity in the future: • Network Monitoring: Deploy continuous network monitoring software to establish traffic baselines and trigger alerts upon detecting abnormal spike patterns.

	• IDS/IPS Deployment: Configure Intrusion Detection Systems (IDS) and Intrusion Prevention Systems (IPS) to automatically analyze and filter out suspicious ICMP traffic. • Log Management: Centralize network and firewall traffic logs to regularly audit inbound traffic and detect unauthorized probing early.
Respond	In the event of a future cybersecurity incident, the team will execute the following response protocol: • Containment: Immediately isolate target entry points and apply emergency blocking rules (e.g., blocking malicious ICMP traffic at the perimeter). • Neutralization: Temporarily take non-critical network services offline to preserve bandwidth and stabilize critical operations. • Analysis: Review firewall, IDS/IPS, and network monitoring logs to analyze the source, vectors, and behavior of the attack. • Process Improvement: Document incident findings, review the incident response plan, and update security controls to patch discovered gaps.
Recover	To safely restore services and return the organization to normal operations: • System Restoration: Bring network services back online in order of priority—restoring critical operational services first, followed by non-critical systems once stability is verified. • Traffic Verification: Monitor network traffic during and immediately following system restoration to confirm normal baseline performance and verify that malicious activity has ceased. • Post-Incident Review: Conduct a post-mortem review with the security team to evaluate the timeline, document lessons learned, and adjust recovery metrics for future continuity planning.

Reflections/Notes:

- **Defense in Depth:** Relying on a single firewall rule isn't enough; combining firewalls, rate limiting, IP verification, and IDS/IPS provides layered security.
- **Proactive Baseline Monitoring:** You can't detect abnormal traffic if you don't know what normal traffic looks like. Network monitoring software is essential for quick detection.
- **Structured Frameworks Work:** Using the NIST CSF ensures that post-incident analysis covers every phase of the lifecycle—from identifying the vulnerability to fully recovering operations.